



# FinishLynx

BUILD SPEC FOR CLAUDE CODE

# FinishLynx Support Portal

*How to build the static troubleshooter site for GitHub Pages*

## READ THIS FIRST

This deck is the spec. The mockup slides show the look. The text slides set the rules. Build the site to match both. Don't invent product features or URLs that aren't in the source files.

# Background

What you're building and why it exists.

## WHO LYNX IS

### Lynx System Developers

Maker of FinishLynx — the photo-finish timing system used at the Olympics, Kentucky Derby, NFL Combine, and tens of thousands of track meets, regattas, and races worldwide. Cameras, software, scoreboards, accessories.

## WHAT THIS SITE IS

### A guided troubleshooter

A static HTML site hosted on GitHub Pages. Users click through a symptom-first decision tree to land on a fix. Linked from a project landing page on the same GitHub repo.

## WHO USES IT

### Timers in the field

Often mid-meet, often mobile, sometimes panicked. They have minutes, not hours. The site has to give the right answer fast — or get them to a real person fast.

# The problem

Lynx has solid documentation. Finding the right doc at the right moment is the hard part.



## Docs exist, but they're scattered

144+ PDFs, QSGs, datasheets, manuals, and YouTube videos across the public site. No single search across all of them. No shared navigation.



## Symptoms aren't filenames

A timer who sees "Error aligning camera" doesn't know to open the Camera Boot Troubleshooting Guide. They search the symptom and hope.



## Support gets the same calls

Recurring issues — RadioLynx start signal, scoreboard scripts, ACM settings — eat support hours that could be reclaimed by a self-serve fix.



## Real use is mobile, mid-meet

The press box, the start line, a folding chair next to a generator. Phone in hand, race in five minutes. Desktop-only doesn't cut it.

# The solution

Three layers: Entry → Decision → Resolution. Plus a search bar for power users and a link library underneath it all.

1

## Entry

User picks a product category from large tiles. One choice cuts the universe by 80%.



2

## Decision

Branching questions narrow the symptom. Each answer routes to the next most useful question.



3

## Resolution

Cause, fix steps, linked docs and videos. A Yes / No / Partially button closes the loop.

### UNDERNEATH IT ALL

Every resolution screen, search result, and category landing page pulls from a single curated link inventory: 144+ PDFs, datasheets, QSGs, manuals, brochures, and videos sourced from finishlynx.com. The list is parsed from Lynx\_Master\_Reference.md at build time.

# Tech stack

Static. Boring. Maintainable. Hosts on GitHub Pages with zero setup.

## USE

### HTML5

Semantic markup. One index.html plus per-category pages OR client-side routing — your call, but keep it simple.

### CSS3

Plain CSS or a single stylesheet. CSS variables for the palette. No SASS, no Tailwind.

### Vanilla JavaScript

Use it for: tree navigation, search filtering, breadcrumb state, feedback button to localStorage. No frameworks.

### JSON data files

Tree structure → tree.json. Resource inventory → resources.json (built from Lynx\_Master\_Reference.md).

### GitHub Pages

Default output: site builds and deploys with no Actions workflow needed. If you add one, keep it minimal.

## DON'T USE

### React / Vue / Svelte

Overkill. Adds a build step, makes the repo harder to maintain, gives no benefit at this scale.

### Tailwind / CSS-in-JS

Plain CSS with variables is sufficient and easier to hand-edit later.

### Backend / database

The site is read-only. No user accounts, no submissions, no server.

### External analytics

No Google Analytics, no Mixpanel, no third-party trackers. Feedback stays in localStorage.

### Heavy dependencies

If a library adds more than a few KB, justify it in a comment or skip it. Vanilla DOM APIs are fine.

Hosting target: a /docs subfolder or gh-pages branch on the same repo where this deck and Lynx\_Master\_Reference.md live. Linked from the project's existing landing page.

# Repo structure

Suggested layout. Adjust if you have a strong reason — but keep it flat and obvious.

```
support/                # GitHub Pages root
├─ index.html           # Landing page
├─ tree.html            # Decision tree pages
├─ search.html          # Search results
├─ browse.html          # Browse all docs
├─ help.html            # "Help me check" pages
├─ support.html         # Talk-to-support
├─
├─ assets/
│   └─ styles.css       # All site CSS
│   └─ app.js           # Tree, search, feedback
│   └─ logo-mark.svg    # Cat-head silhouette
│   └─ icons/           # Inline SVG icons
├─
├─ data/
│   └─ tree.json        # Decision tree data
│   └─ resources.json   # 144+ docs (parsed)
├─
└─ scripts/
    └─ build-resources.js # MD parser → JSON
```

## KEY PRINCIPLES

### ● Flat over deep

Don't create nested `src/components/ui/forms/`. Maintainers should see every file at a glance.

### ● Data is the product

Site logic is small. `tree.json` and `resources.json` are the real content. The HTML/JS is a renderer.

### ● One source of truth

`resources.json` is rebuilt from `Lynx_Master_Reference.md`. Never hand-edit it.

### ● Editable by humans

`tree.json` should be readable and editable in any text editor. No magic numbers, no codes.

### ● Hash-routable

Each tree node has a stable URL like `#/camera/error-aligning`. Shareable, bookmarkable.

# Source of truth: Lynx\_Master\_Reference.md

The link inventory lives here. Parse it once at build time. Never hand-edit the JSON.

## WHAT THE SOURCE LOOKS LIKE

```
### 1. Lynx-Sold Cameras

#### Datasheets

##### EtherLynx Vision PRO X Datasheet (5L700)
- **URL:** https://finishlynx.com/wp-content/...
- **File Status:** In Folder
- **Use:** Product specs and features
  for the EtherLynx Vision PRO X (5L700).

##### EtherLynx 2000+ Datasheet (5L420)
- **URL:** ❌
- **File Status:** In Folder
```

## WHAT IT BECOMES

```
// data/resources.json
[
  {
    "title": "EtherLynx Vision PRO X
              Datasheet (5L700)",
    "url": "https://finishlynx.com/...",
    "category": "Lynx-Sold Cameras",
    "subcategory": "Datasheets",
    "use": "Product specs and features",
    "type": "datasheet"
  },
  // entries with URL: ❌ are SKIPPED
]
```

## PARSING RULES

1. ### N. Title → category. Drop the leading number.
2. ##### Title → subcategory.
3. ##### Title → resource entry.
4. Capture URL, File Status, Use as fields beneath each ##### heading.
5. If URL is ❌, skip the entry — never invent a URL.
6. Infer type from subcategory name: "Datasheets" → datasheet, "QSGs" → qsg, "Manuals" → manual, "Videos" → video, etc.
7. Output one flat array. Sorting and grouping happens in the renderer, not the data file.

## PART TWO

# What each page should look like

*The next nine slides are visual mockups for every page in the site. Match the layout, spacing, color, typography, and copy. Treat the mockups as a contract.*

### PAGES TO BUILD

Landing · Symptom selector · Diagnostic question · Resolution · Search results · Browse all docs · Tree as data · Mobile · Help me check · Talk to support



# Landing page · Entry layer

First screen the user sees. Search bar above the fold for power users; tile grid for symptom-first navigation.

support.finishlynx.com

 **FinishLynx**

Support

SearchBrowse DocsContact Support

# What's happening?

Pick a category to start. Or search if you already know what you're looking for.

 Search known fixes, docs, and videos...

Search

**Camera**

EtherLynx Vision, PRO, PRO X

**FinishLynx Software**

Capture, results, scoreboards

**Scoreboard / Display**

ResultTV, VDM, LED panels

**Wind Gauge**

Lynx Wind, controller, cable

**RadioLynx / Wireless**

Start sensor, signal loss

**ReacTime**

False start, calibration

**Transponder Timing**

MYLAPS, IPICO, BibTag

**Networking / Setup**

IPs, switches, firewall

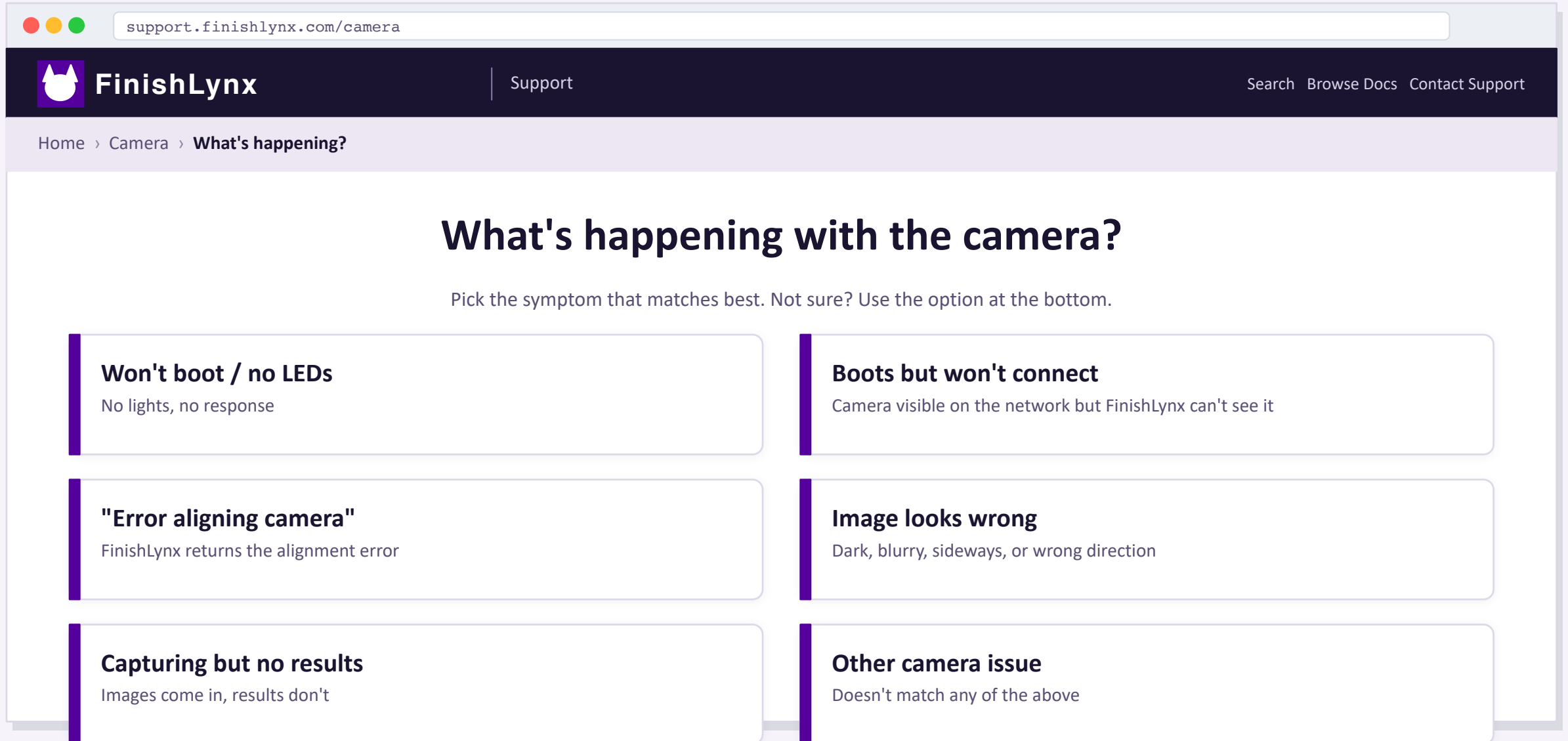
**Something else**

Search or contact support

Mockup 01 · Landing page

## Second screen · Symptom selector

After picking a category, the user picks a symptom. Breadcrumb appears and stays clickable for the rest of the session.



The screenshot shows a web browser window with the URL `support.finishlynx.com/camera`. The page has a dark header with the FinishLynx logo and navigation links. A breadcrumb trail is visible below the header. The main content area is titled "What's happening with the camera?" and includes a sub-instruction. Below this, there are six selectable symptom cards arranged in a 3x2 grid.

support.finishlynx.com/camera

**FinishLynx** | Support

Search Browse Docs Contact Support

Home › Camera › **What's happening?**

### What's happening with the camera?

Pick the symptom that matches best. Not sure? Use the option at the bottom.

|                                                                          |                                                                                             |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Won't boot / no LEDs</b><br>No lights, no response                    | <b>Boots but won't connect</b><br>Camera visible on the network but FinishLynx can't see it |
| <b>"Error aligning camera"</b><br>FinishLynx returns the alignment error | <b>Image looks wrong</b><br>Dark, blurry, sideways, or wrong direction                      |
| <b>Capturing but no results</b><br>Images come in, results don't         | <b>Other camera issue</b><br>Doesn't match any of the above                                 |

# Diagnostic question · Visual answers

Visual aids are inline, not behind links. Each LED state is its own answer.

support.finishlynx.com/camera/error-aligning

 **FinishLynx**

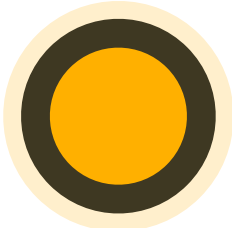
Support

Search | Browse Docs | Contact Support

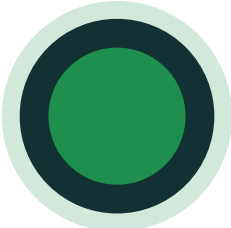
Home › Camera › Error aligning camera › LED check

## What color are the camera LEDs right now?

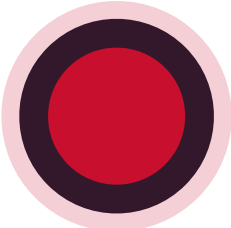
Look at the back panel. Pick the state that matches what you see.



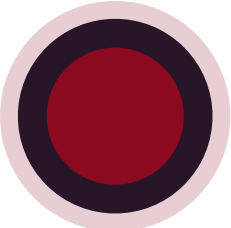
**Flashing amber**  
Waiting for FinishLynx



**Solid green**  
Powered on, not booting



**Flashing red**  
Booting (normal)



**Solid red >1 min**  
Boot failure

**No lights at all**

**I can't tell — show me how to check**

Mockup 03 · Diagnostic question with visual answers

# Resolution screen · Where the site earns its keep

Cause + numbered fix steps + linked references + the all-important Yes / No / Partially feedback row.

support.finishlynx.com/fix/camera-boot-failure

FinishLynx

Support

SearchBrowse DocsContact Support

Home › Camera › Error aligning camera › Solid red >1 min › **Fix**

LIKELY CAUSE

Camera failed to boot. "Error aligning camera" is always a boot issue.

Try this

1

Power-cycle the camera. Unplug, wait 10 seconds, plug back in.

2

Watch the LEDs through the full boot sequence (~45–60 seconds).

3

If still solid red, swap the Ethernet cable between the PoE injector and camera.

4

If still failing, try a different PoE injector or known-good port on the switch.

5

If LEDs never reach flashing amber, follow the BIOS chip replacement guide.

REFERENCES

DOC

Camera Boot Troubleshooting Guide

DOC

Camera BIOS Chip Replacement Guide

VID

Diagnosing camera boot failures (8 min)

FIX

Known Fix: Solid red LED, no boot

Did this fix it?

Your answer trains the system.

Yes

No

Partially

Mockup 04 · Resolution screen with feedback loop

# Search results · Power-user path

Results in priority order: Known Fixes first, then docs, then videos, then guided paths.

support.finishlynx.com/search?q=vdm+invalid+code

 **FinishLynx**

Support

SearchBrowse DocsContact Support

 vdm invalid code

Search

12 results

All

Known Fixes (3)

Docs (5)

Videos (2)

Guided paths (2)

FIX

**VDM "Invalid code" appears even when the code worked**  
Restart the camera and FinishLynx before concluding the code failed. Verify Camera [#] (HDMI) in Scoreboard > Options.  
*Confirmed Fix · Updated this month*

FIX

**ResultTV-VDM not displaying after upgrade**  
Check that the camera's HDMI scoreboard entry is enabled in FinishLynx scoreboard options after applying the upgrade code.  
*Confirmed Fix*

DOC

**ResultTV-VDM Quick Start Guide**  
Full setup procedure including upgrade code activation and HDMI scoreboard configuration in FinishLynx.  
*PDF · Quick Start Guide*

VID

**Activating a VDM upgrade code (Tutorial)**  
5-minute walkthrough showing the upgrade code entry, restart, and verification steps.  
*YouTube · Tutorial Videos*

# Browse all docs · Where the link inventory pays off

Categorized, filterable view of every resource in resources.json. This page alone uses 100+ of the URLs.

support.finishlynx.com/browse

FinishLynx

Support

SearchBrowse DocsContact Support

Home › Browse all docs

Filter docs by keyword, category, or type...

CATEGORY

Cameras26

Timing accessories31

Software18

Domestic packages16

International packages22

Third-party integration14

Scoreboards / displays17

TYPE

☒ Datasheets

☒ QSGs

☒ Manuals

☐ Videos

Showing 144 docs across 7 categories

DOC

EtherLynx Vision PRO X Datasheet (5L700)

Cameras › Datasheets · Product specs and features

QSG

FieldLynx QSG

Software › QSGs · Setup and operation of full FieldLynx field events system

MAN

LynxPad User Guide

Software › Manuals · Full reference for LynxPad meet management software

QSG

RadioLynx Wireless Start Setup QSG

Timing accessories › QSGs · Wireless start signal pairing and operation

DOC

MYLAPS BibTag System Manual

Third-party › Manuals · Full reference for MYLAPS BibTag chip timing

VID

Camera Boot Troubleshooting walkthrough

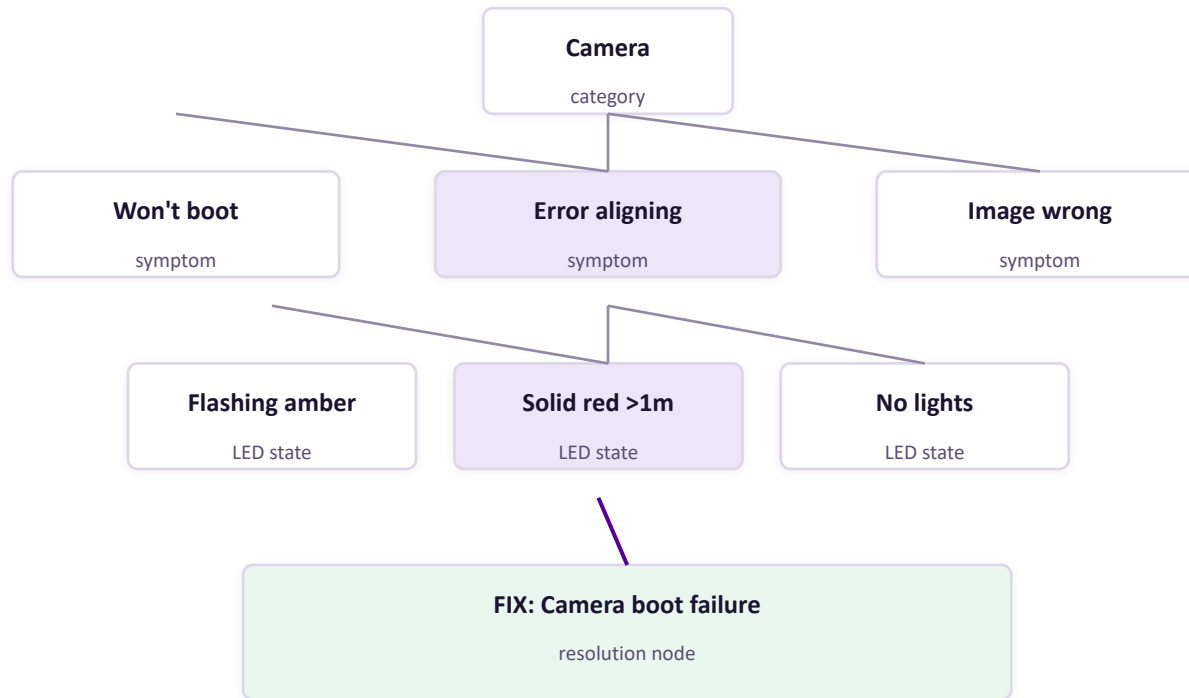
Cameras › Videos · Diagnosing LED states and boot failures

Mockup 06 · Browse all docs (uses full link inventory)

# Under the hood · Tree as JSON, not code

Every branch and resolution lives as data. New Known Fixes are added by appending nodes — no front-end edits required.

## Decision tree (visualized)



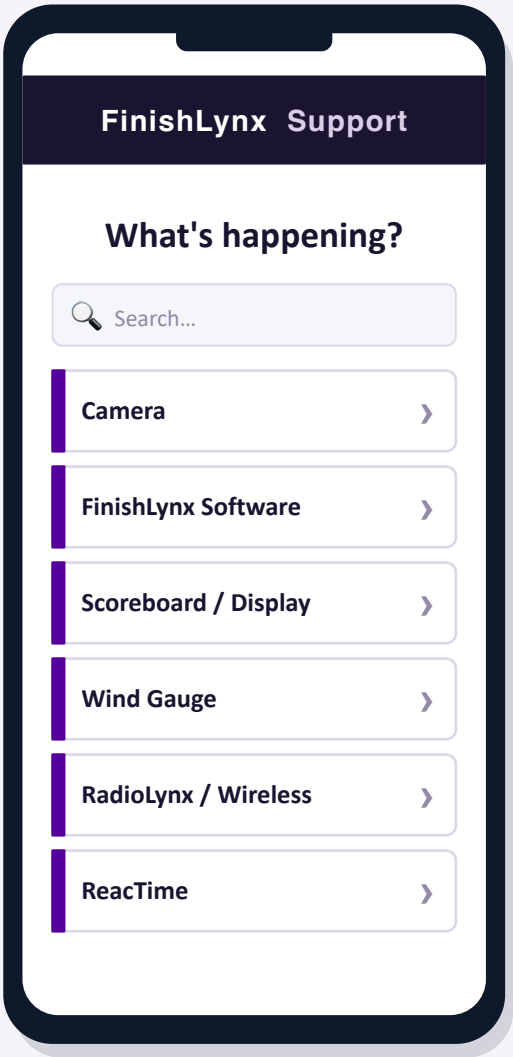
*Multiple branches can converge on the same resolution node — happens constantly in real troubleshooting.*

## node definition · camera\_error\_aligning.json

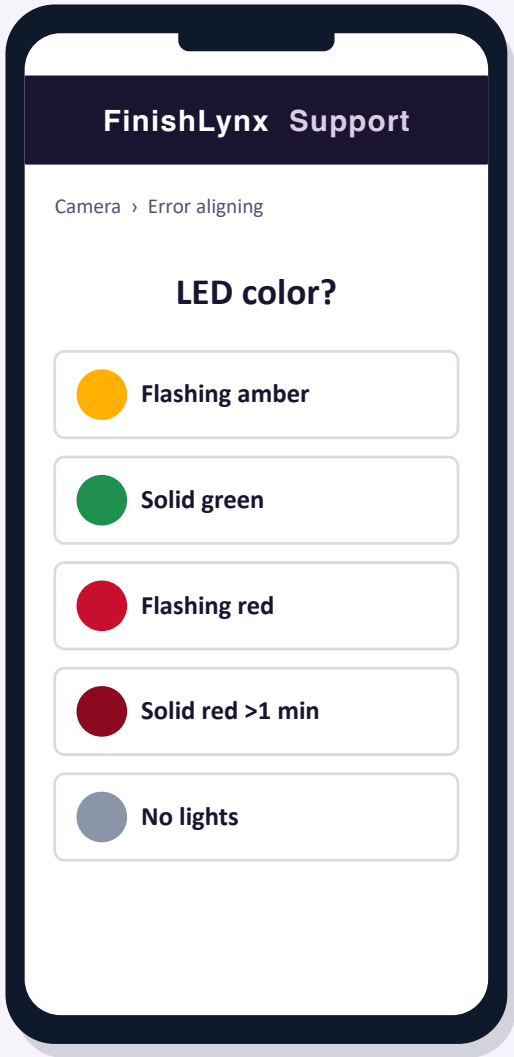
```
{
  "id": "camera_error_aligning",
  "question": "What color are the LEDs?",
  "options": [
    { "label": "Flashing amber", "next": "cam_waiting" },
    { "label": "Solid red >1m", "next": "cam_boot_fail" },
    { "label": "No lights", "next": "cam_power_check" }
  ],
  "context_tags": ["camera", "boot"]
}
```

# Mobile view · Press box, mid-meet

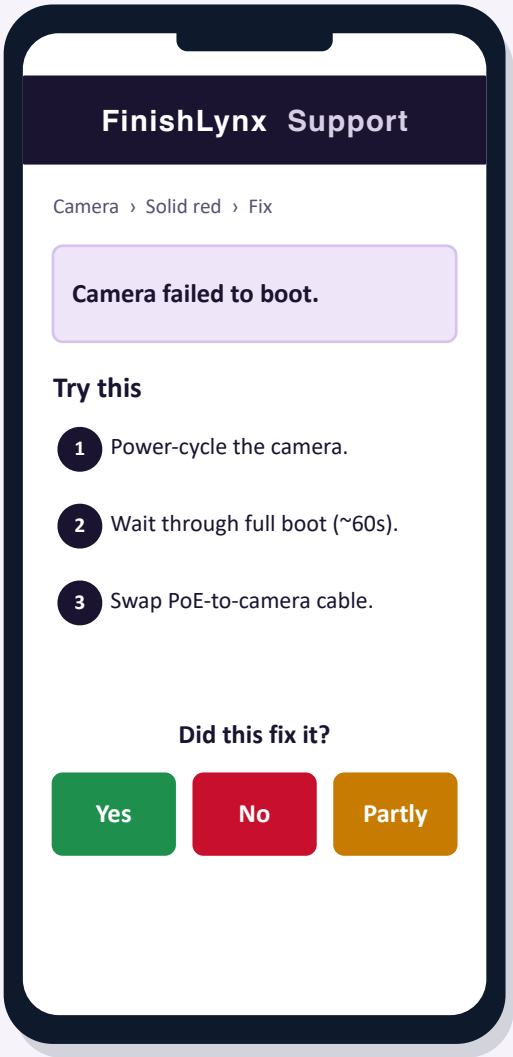
The most common real-world use: phone in hand, race in 5 minutes. Same content, single-column layout, big tap targets.



Landing



Question



Resolution



# "Help me check" page · Clarification mode

When the user picks "I can't tell", the site explains how to look — then drops them right back at the original question.

support.finishlynx.com/help/find-camera-leds

 **FinishLynx**

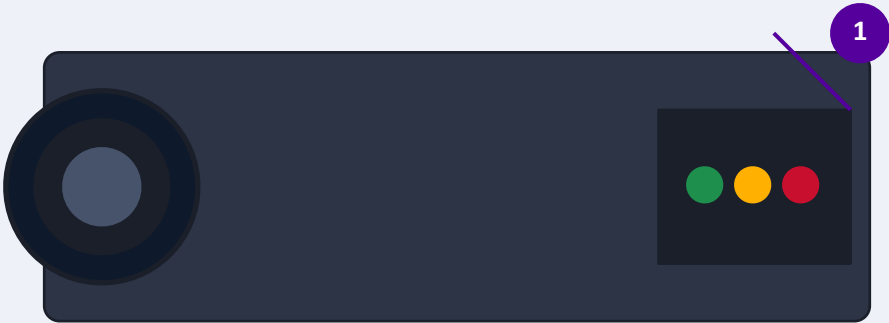
Support

Search Browse Docs Contact Support

Home › Camera › Error aligning camera › LED check › **Help**

## Where to find the camera LEDs

Diagram



Camera back panel — LEDs on the right

### How to find them

- 1 Find the back panel of the camera (opposite the lens).
- 2 Look for a recessed black panel on the right side.
- 3 Inside that panel are three indicator LEDs stacked vertically.
- 4 The bottom LED is the boot status — the one this question is asking about.

Got it — back to the question ›

# Escalation page · Talk to support

Last-resort path. TeamViewer download first, support page second — never just a phone number.

support.finishlynx.com/talk-to-support

FinishLynx

Support

SearchBrowse DocsContact Support

Home › Camera › Error aligning camera › **Talk to support**

Let's get you connected to a person

*Before you call: install TeamViewer so support can see what you're seeing. It's the fastest path to a fix.*

STEP 1

Install TeamViewer

Support uses TeamViewer to see your screen and walk through the issue with you. Install it before you call so the session can start right away.

Download from finishlynx.com/support

STEP 2

Then visit the support page

Phone numbers, hours, and contact options are all listed on the support page. Don't lose your spot — open it in a new tab.

Open www.finishlynx.com/support

Tell support what you've already tried: Camera › Error aligning › Solid red >1 min › Steps 1–3 attempted, still failing

Mockup 10 · Escalation page

# Build plan

Suggested phases. Ship a working slice early; expand from there.

## Phase 1 Plumbing

Set up repo, scripts/build-resources.js, generate resources.json from Lynx\_Master\_Reference.md.  
Verify all 144 entries (minus **X** skips) are captured.

## Phase 2 Skeleton

index.html, support.html, browse.html. Header, footer, breadcrumb component, palette, typography. No real content yet — wire up the navigation shell.

## Phase 3 Browse-all-docs page

Render every entry in resources.json as a categorized, filterable list. This is the highest-leverage page — it makes the link inventory pay off immediately.

## Phase 4 Decision tree (one branch deep)

Build tree.json with the Camera → Error aligning → LED state → Resolution branch fully wired. This is the reference implementation for every other branch.

## Phase 5 Search

Client-side search across resources.json + tree.json. Highlight matches. Filter by type. No external search service.

## Phase 6 Other category landing tiles

Each remaining category (Software, Scoreboards, RadioLynx, etc.) gets a tile that opens a curated link list pulled from resources.json. Build a deep tree only when content is ready.

## Phase 7 Mobile pass

Single-column layout below 720px. Tap targets ≥44px. Test the full flow on a phone before shipping.

## Phase 8 Feedback + polish

Yes/No/Partially writes to localStorage with timestamp + node id. Add the Help-me-check pages and the Talk-to-support escalation.

# Hard rules from Lynx\_Master\_Reference.md

These exact statements appear in real fix content. Do not paraphrase them. Do not soften them. Do not invent new ones.

|                                                                                                                                                             |                                                                                                                                                                        |                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <p><b>FIREWALL</b></p> <p>Always use C:\Lynx\Lynx.exe when adding FinishLynx to Windows Firewall. Steps go through Windows Security, not Control Panel.</p> | <p><b>CAMERA BOOT</b></p> <p>"Error aligning camera" is always caused by the camera not booting correctly. Always reference the Camera Boot Troubleshooting Guide.</p> | <p><b>LENS ADAPTERS</b></p> <p>C-Mount lenses require the CS-C 5mm ring adapter. CS-Mount lenses must NOT use the adapter.</p> |
| <p><b>FINISHLYNX LITE</b></p> <p>Installs to C:\LynxLite, not C:\Lynx. Does not support meet management database connections.</p>                           | <p><b>RADIOLYNX</b></p> <p>Always set Wired Start Sensor to None on the Inputs tab first.</p>                                                                          | <p><b>ACM</b></p> <p>Uncheck "Allow button override" in File &gt; Options &gt; General.</p>                                    |
| <p><b>SCOREBOARD STATUS</b></p> <p>"Running" only means FinishLynx is sending data — not that the scoreboard is receiving it.</p>                           | <p><b>CAMERA IPs</b></p> <p>Auto-assigned by FinishLynx — never manually configured by the user.</p>                                                                   | <p><b>MEET DIRECTORY</b></p> <p>C:\Meets</p>                                                                                   |
| <p><b>FINISHLYNX API</b></p> <p>FinishLynx has no API. Third-party integrations use .LIF file export or scoreboard output via .LSS scripts.</p>             | <p><b>SUPPORT PAGE</b></p> <p>Always include the full URL: <a href="http://www.finishlynx.com/support">www.finishlynx.com/support</a></p>                              | <p><b>TEAMVIEWER</b></p> <p>Direct customers to download TeamViewer first before calling support.</p>                          |

# Visual design system

Brand-aligned with the official FinishLynx style guide. Use exactly these values.

## COLORS

|               |             |              |                |                 |            |                  |                |
|---------------|-------------|--------------|----------------|-----------------|------------|------------------|----------------|
| #55009C       | #3A006B     | #EFE5F8      | #1A1430        | #F6F4FA         | #1F8F4E    | #C77B00          | #C8102E        |
| Primary brand | Hover state | Callouts     | Text + headers | Page background | Yes button | Partially button | No button only |
| Lynx Purple   | Purple Dark | Purple Light | Ink            | BG              | Success    | Warn             | Alert          |

## T Y P O G R A P H Y

### FinishLynx

Logo wordmark · Eras Demi ITC · Finish regular + Lynx bold

Headers and body · System UI / Calibri / Arial fallback

Code / monospace · Consolas, Menlo, monospace

### P R O D U C T N A M I N G ( N E V E R D E V I A T E )

FinishLynx · LynxPad · EtherLynx · ReactTime · Identilynx · ResultTV · FieldLynx · RadioLynx · VeriLynx · LaserLynx · AirLynx

Capital F, capital L. CamelCase compounds. No spaces, no hyphens, no "finishlynx".

## L O G O M A R K

Use the official FinishLynx cat-head silhouette. The SVG should live at assets/logo-mark.svg. Source: finishlynx.com — download lynx-head-512-200x200.png or the FinishLynx vector logo (.eps). Do not invent or substitute a different mark.

# Guardrails

Things to never do, even if it would make the site "better."

## Never invent product features



If a capability isn't in `Lynx_Master_Reference.md`, `FinishLynx_Known_Fixes.md`, or the official QSGs, it does not exist for the purposes of this site.

## Never invent URLs



Every link must come from `resources.json` (parsed from the Master Reference). If a doc doesn't have a URL there, don't link it. Don't construct one from a pattern.

## Use exact hard-rule wording



The hard rules slide lists them verbatim. Don't paraphrase "FinishLynx has no API" into something softer. The wording is load-bearing.

## Don't fabricate Confirmed Fix status



`FinishLynx_Known_Fixes.md` distinguishes Confirmed Fix from PENDING. Preserve that distinction in any UI label.

## Don't paraphrase product names



FinishLynx, LynxPad, EtherLynx, etc. are exact compounds. Never "finish-lynx" or "Finish Lynx" or "finishlynx".

## Skip entries silently



Master Reference entries with URL:  are not yet linkable. Don't render them as broken links. Don't render them at all.

## Test empty states



What happens when search returns nothing? When a tree branch has no resources? Don't ship without handling these.

## Mobile is not optional



Real users are on phones in noisy press boxes. If it's painful on a 380px viewport, the site has failed.

## Ask when unsure



If a spec is ambiguous, leave a TODO comment rather than guessing. A clear TODO is easier to fix than invented behavior.

## Commit `resources.json`



Even though it's generated, commit it so GitHub Pages serves it without a build step. Re-run `scripts/build-resources.js` whenever the Master Reference changes.

# Out of scope for v1

Don't build these now. They're tracked here so they don't drift in by accident.

V2

Admin / editor view

A dashboard for editing tree nodes, viewing feedback stats, and promoting PENDING fixes to Confirmed. Useful eventually — overkill for a static site.

V2

Full decision tree for every category

v1 builds Camera > Error aligning end-to-end as a reference implementation. Other categories get curated link lists. Build the full tree out as content matures.

V2

Localization

English only for v1. The Master Reference and source docs are English-only at this stage.

LATER

Analytics dashboard

Page views, top branches, drop-off points. Worth measuring once there's a backend, but not now.

V2

Server-side feedback collection

A real backend that captures Yes/No/Partially clicks across all visitors. v1 stores feedback in localStorage only. Add a serverless endpoint when you're ready.

V2

Authentication / personalization

No user accounts, no saved sessions, no role-based content. The site is public and anonymous.

LATER

Chatbot / conversational UI

The same tree.json could feed an LLM-driven chat interface later. Build the click-through version first — it's debuggable and the content stays the same.

LATER

Embed-able support widget

A floating chat-style trigger that opens the troubleshooter on any page of finishlynx.com. Out of scope until v1 stabilizes.

## DEFINITION OF DONE

# Ship checklist

*Every item below is checkable before pushing to GitHub Pages.*

- ☐ scripts/build-resources.js parses Lynx\_Master\_Reference.md and writes data/resources.json with no errors
- ☐ Every entry in resources.json has a non-empty URL — no **×** entries leaked through
- ☐ Landing page tiles open the right category browse pages
- ☐ Camera > Error aligning > LED state > Solid red > 1m branch lands on the boot-failure resolution
- ☐ Resolution screen shows numbered steps, linked references, and a working Yes/No/Partially feedback row
- ☐ Feedback writes to localStorage with timestamp + node id; visible in DevTools
- ☐ Search across resources.json and tree.json returns matches within ~50ms
- ☐ Browse all docs page renders 130+ entries (full inventory minus skipped) with category and type filters
- ☐ Mobile layout works on a 380px viewport: single column, tap targets  $\geq 44$ px, no horizontal scroll
- ☐ Help-me-check page returns the user to the question they came from (URL hash preserved)
- ☐ Talk-to-support page links to [www.finishlynx.com/support](https://www.finishlynx.com/support) and mentions TeamViewer download first
- ☐ Site loads on GitHub Pages with no console errors and no broken images